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import cv2 # Import OpenCV for image processing
import numpy as np # Import NumPy for matrix operations
import matplotlib.pyplot as plt # Import Matplotlib for visualization

# Step 1: Create a synthetic static background
background = np.zeros((200, 200), dtype=np.uint8) # Create a blank black image

# Step 2: Create a surveillance frame with a moving object
frame = background.copy()
cv2.rectangle(frame, (70, 70), (120, 120), 255, -1) # Draw a white square representing a moving object

# Step 3: Detect the moving object using frame difference
motion = cv2.absdiff(background, frame) # Compute absolute difference between frames

# Step 4: Convert the difference into a binary motion mask
_, motion_mask = cv2.threshold(motion, 50, 255, cv2.THRESH_BINARY)

# Step 5: Display the images
plt.figure(figsize=(9,3))

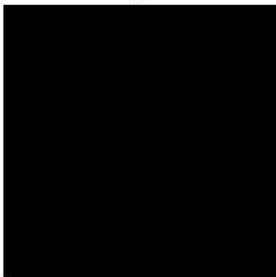
plt.subplot(131)
plt.imshow(background, cmap='gray')
plt.title("Background")
plt.axis("off")

plt.subplot(132)
plt.imshow(frame, cmap='gray')
plt.title("Surveillance Frame")
plt.axis("off")

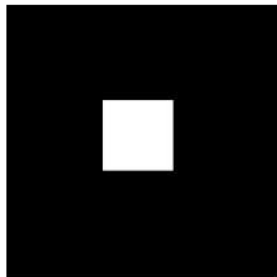
plt.subplot(133)
plt.imshow(motion_mask, cmap='gray')
plt.title("Detected Motion")
plt.axis("off")

plt.show()
```

Background



Surveillance Frame



Detected Motion

